



VPixx Technologies
Vision Science Solutions

VideoBahn™
The NEW fast road to SUPERIOR data

VIEWPixx



DIGITAL, ANALOG AND AUDIO I/O

- Button box interface
- 24 TTL trigger inputs and outputs
- Stereo audio input and output
- Analog inputs and outputs

SYNCHRONIZED DATA ACQUISITION

All digital, analog, and audio inputs and outputs feature microsecond synchronization to video refresh.

VIDEO

- 12-bit RGB intensity
- 1920 x 1080 resolution at 120 Hz
- Scanning LED backlight with direct RGB LED array

120 Hz CRT Replacement



OVERVIEW

The VIEWPixx is a complete display toolbox which has been conceived specifically to replace CRTs in vision science labs. The VIEWPixx features high-performance industrial LCD glass and a panel controller specifically designed to support vision research. Our innovative LED backlight design features superior display uniformity, and a wide color gamut exceeding that of any CRT. In addition, the VIEWPixx includes an array of peripherals which often need to be synchronized to video during an experiment, including a stereo audio stimulator, a button box port for precise reaction-time measurement, triggers for electrophysiology equipment, and even a complete analog I/O subsystem. Because we implemented the video controller and peripheral control on the same circuit board, you can now successfully synchronize all of your subject I/O to video refresh with microsecond precision.

CONTACT US

Phone : (514) 328-7499

1 (844) 488-7499 - Toll Free USA/Canada

Web : vpixx.com

VPixx Technologies Inc.
630 Clairevue West, suite 301
Saint-Bruno, QC Canada, J3V 6B4

SPECIFICATIONS

LCD SPECIFICATIONS

- Display resolution: 1920(H) x 1200(V) pixels
- 22.5-inch display size (diagonal)
- Pixel pitch: 0.252(H) x 0.252(V) mm
- Pixel arrangement: RGB (Red dot, Green dot, Blue dot) vertical strip
- Active matrix LCD
- 12 bits of resolution on each of the RGB channels
- 100 Hz to 120 Hz refresh rate with zero latency stimulus presentation
- 60 Hz to 100 Hz refresh rate with internal frame buffering
- Grey-to-Grey response time:
 - 1 ms typical in scanning backlight mode
 - 7 ms typical in normal backlight mode
- Luminance:
 - 100 cd/m2 in scanning backlight mode
 - 250 cd/m2 in standard backlight mode
- Uniformity: 95% over 95% of display area
- Contrast ratio: typical 800:1
- Viewing angle: 176° (Horizontal), 176° (Vertical)
- Polarizer surface: Antiglare

BACKLIGHT SPECIFICATIONS

- Scanning LED backlight
- Direct RGB LED array
- Wide gamut LED
- Factory white point D65

VIDEO PROCESSING

- Video input: 1920 x 1200 pixels, 60 to 120 Hz, 24 bits (Dual link DVI)
- Deterministic timing between reception of video signal and update of display pixels
- Completely bypass all image processing "enhancements" prevalent in standard consumer LCD panels
- Multiple displays can be synchronized, showing copies or subsets of original video

POWER

- Power consumption: 180 W
- Input voltage: 48 VDC - 3.75 A
- International AC adaptor input: 90 VAC - 264 VAC (47 Hz - 63 Hz)

VIEWPixx Stand

- Mounting standards: VESA MIS-D/E, MIS-F
- Hole pattern: 100 x 100 mm & 75 x 75 mm

Lift	Tilt	Pan	Rotation	VESA
5" 13 cm	30°	Base 360°	90° P/L	MIS-D/E MIS-F

ANALOG TO DIGITAL CONVERTER***

- Number of channels: 16 (or 8 differential), on DB-25 connector
- Converter resolution: 16 bits
- Maximum sampling rate: 200 kSPS per channel
- Frequency programming modes:
 - Samples per second
 - Samples per video frame
 - Nanoseconds per sample
- Simultaneous sampling across all channels
- Input range: ± 10 V
- Input impedance: $1.6 \times 10^8 \Omega // 3$ pF
- Absolute maximum input tolerance ± 12 V

DIGITAL TO ANALOG CONVERTER***

- Number of channels: 4 on DB-25 connector
- Converter resolution: 16 bits
- Maximum sampling rate: 1 MSPS per channel
- Frequency programming modes:
 - Samples per second
 - Samples per video frame
 - Nanoseconds per sample
- Simultaneous output updates
- Output range: ± 10 V
- Drive capability: ± 25 mA, 250 mW per channel

AUDIO CODEC***

- Audio line in, microphone in, speaker out, on 3.5 mm jacks
- Stereo microphone input amplifier resistance: 20 k Ω
- Microphone sampling rate: 96 kHz
- Programmable microphone bias voltage range: 2.0 V to 3.1 V
- Stereo DAC sampling rate 96 kHz

DIGITAL INPUT

- Number of digital inputs: 24 on DB-25 connector
- Input termination: > 20 k Ω pullup to 3.3 V
- Input tolerance: 5 V

DIGITAL OUTPUT

- Number of digital outputs: 24 on DB-25 connector
- Output drive stage: 5 V through 25 k Ω series resistor
- Maximum output current:
 - Source: 15 mA
 - Sink: 12 mA

SOFTWARE

Software support includes a low-level ANSI C API, MATLAB/Octave and Python libraries for use under Mac OS X, Microsoft Windows, and Linux.



***These functionalities are available only with the VIEWPixx full version (VPX-VPX-2001C)

ORDERING INFORMATION

Description: VIEWPixx Full LCD display and data acquisition system

P/N: VPX-VPX-2001C

Description: VIEWPixx Lite LCD display and data acquisition system

P/N: VPX-VPX-2000A

